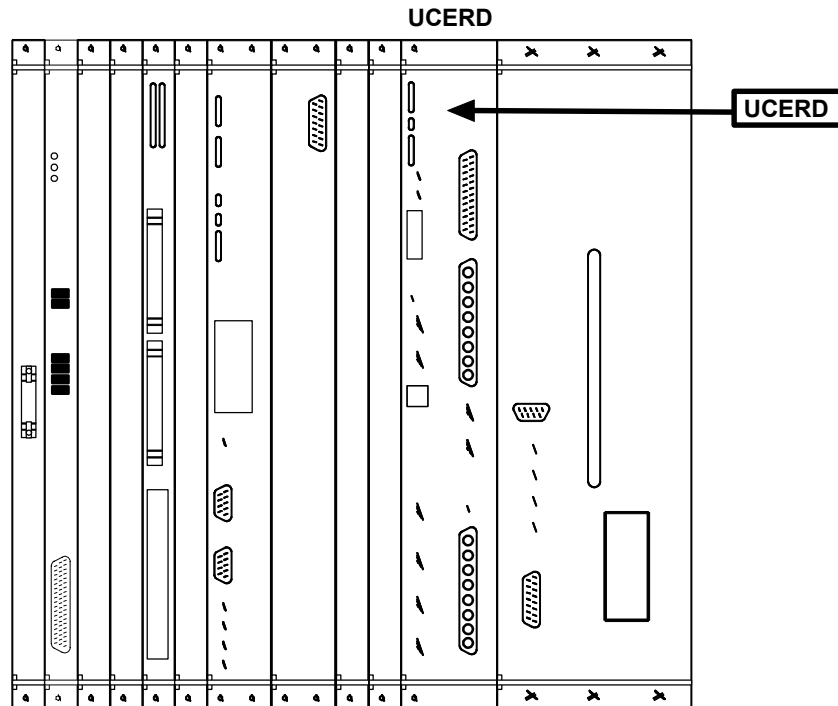


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1- UCERD ASSEMBLY MODULE

The UCERD (Universal Combined Exciter/Receiver Digital) Module is located in slots 13,14, and 15 of the ISE/TPS Chassis. Refer to Illustration 1-1 for the UCERD Assembly.



UCERD ASSEMBLY LOCATION
ILLUSTRATION 1-1

1-1 Removal of UCERD Assembly

1. Remove the System Cabinet front cover.
2. Turn off the main circuit breaker, CB1 on the front of the system cabinet.



Equipment damage possibility. Loosen the two front screws from the blank panel to the left (slot # 12) of the UCERD Assembly and remove the blank panel. Components and RF fingers (EMC) on the UCERD can be damaged if the blank panel is not removed.



Equipment damage possibility. ESD precautions must be used when removing the UCERD assembly or replacing any modules on the UCERD.

3. Remove cable connections J101, J108, and J109 from the front of the UCERD.

- Loosen the two front panel screws. Remove the UCERD Assembly from the ISE/TPS Chassis and place on an ESD Static mat for further disassembly.

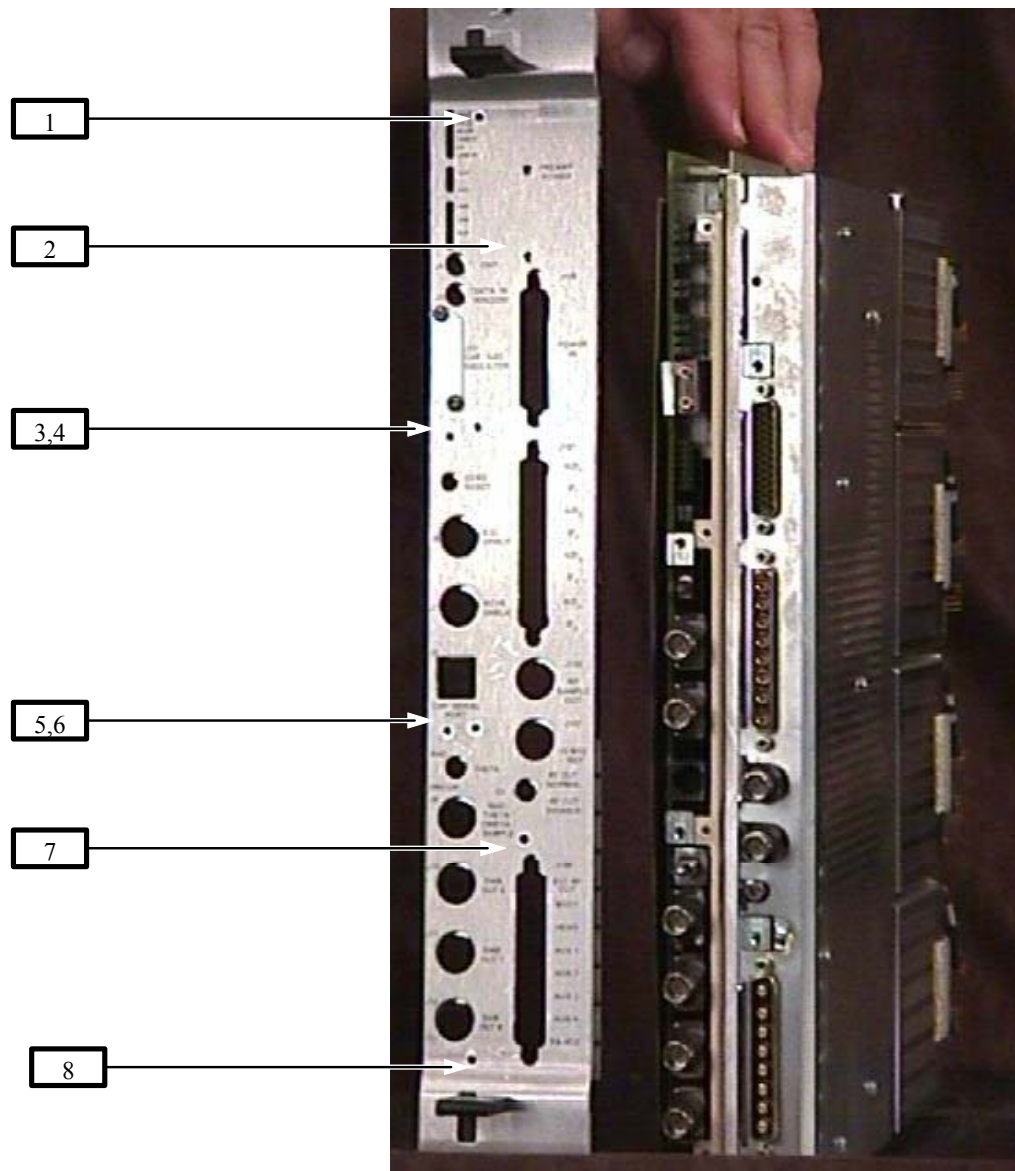
1-2 Removal and Replacement of Front Face Plate



THE UCERD MUST BE DISASSEMBLED IN THE PRESCRIBED ORDER OR IT WILL BE DAMAGED

1-2-1 Removal of Face Plate (Not required for Receiver replacement)

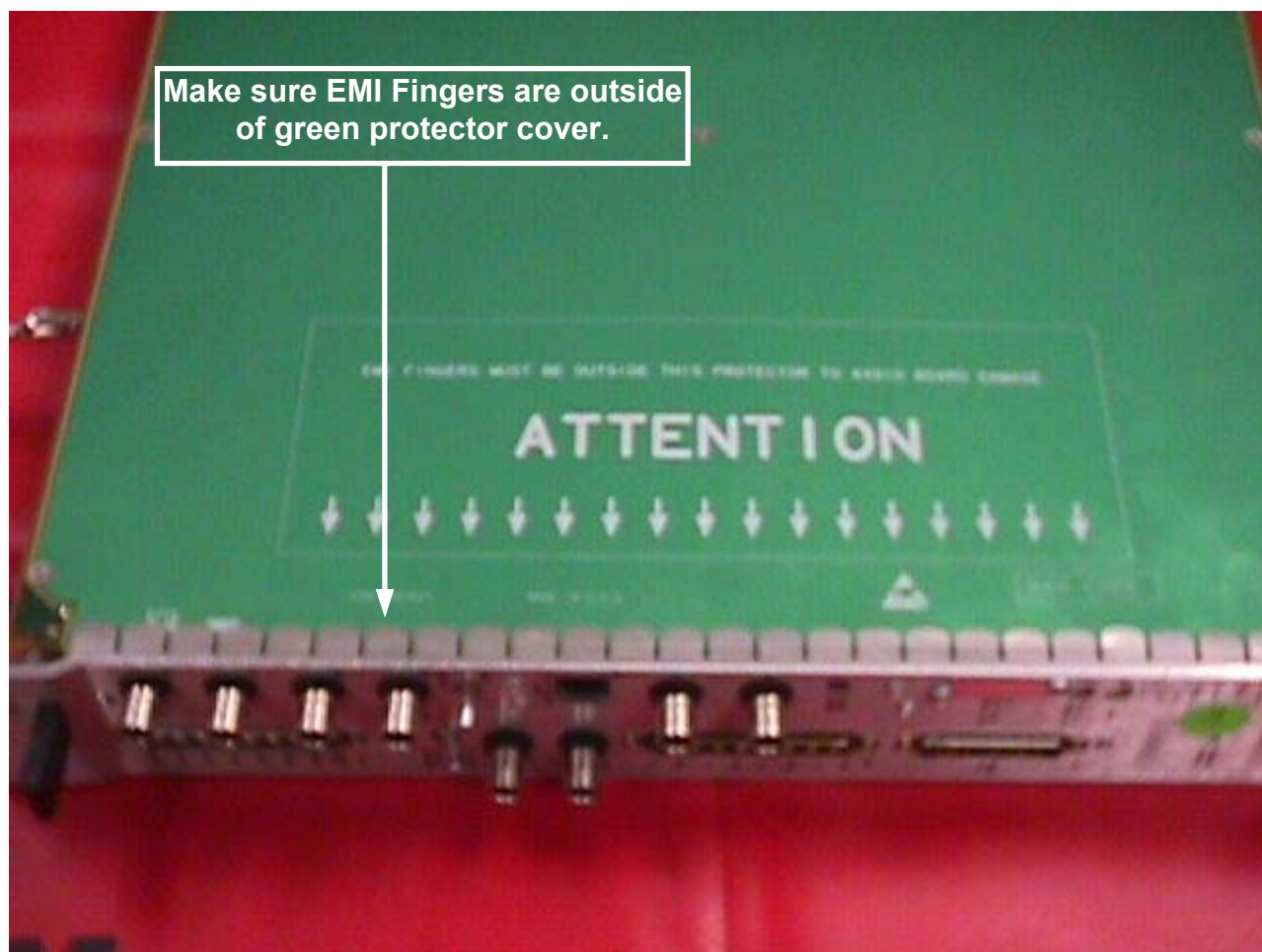
- Make sure the UCERD is on a firm countertop with an ESD mat and a static strap.
- Remove the 8 screws that attach the front face plate. See Illustration 1-2.



UCERD FRONT COVER FACE PLATE
ILLUSTRATION 1-2

1-2-2 Replacement of Face Plate

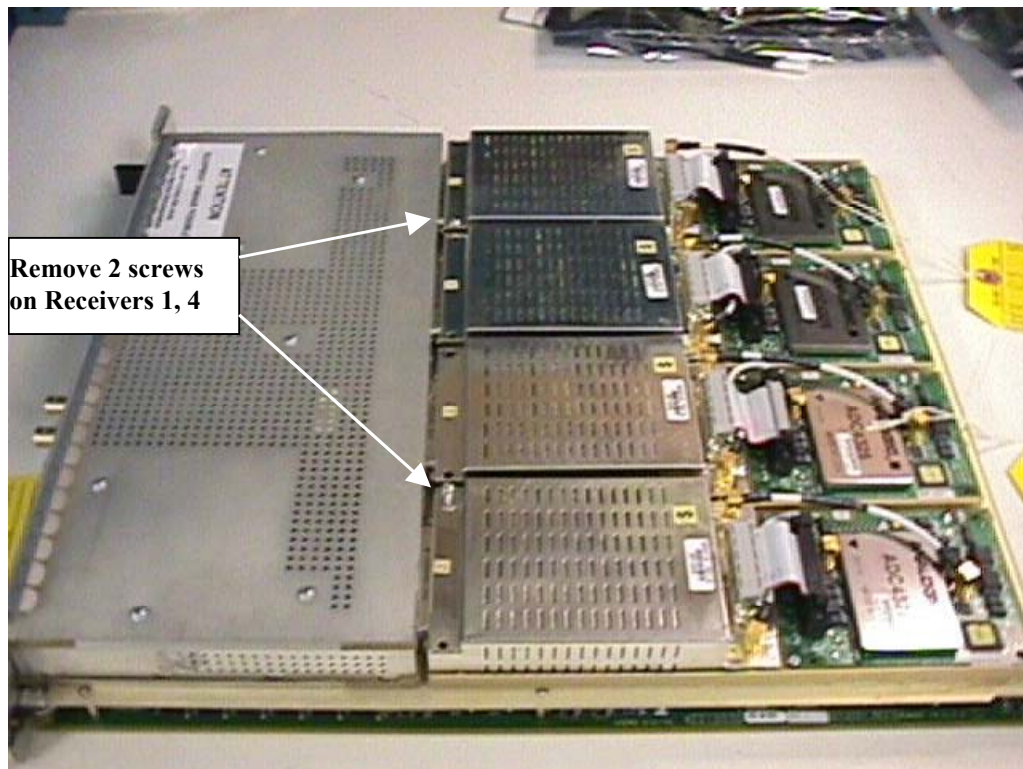
1. For replacement, it is critical that the EMI fingers be on the outside of the green cover or serious board damage will result. See Illustration 1-3.
2. Install the 8 screws as shown in Illustration 1-2. Install all 8 screws loosely, then tighten.



REPLACEMENT OF UCERD FACE PLATE
ILLUSTRATION 1-3

1-3 Replacement of Exciter Module

1. Remove the two screws on Receiver 1 and 4 near the Exciter. See illustration 1-4. **Do NOT remove screws from the Exciter cover as there are no serviceable parts inside.**

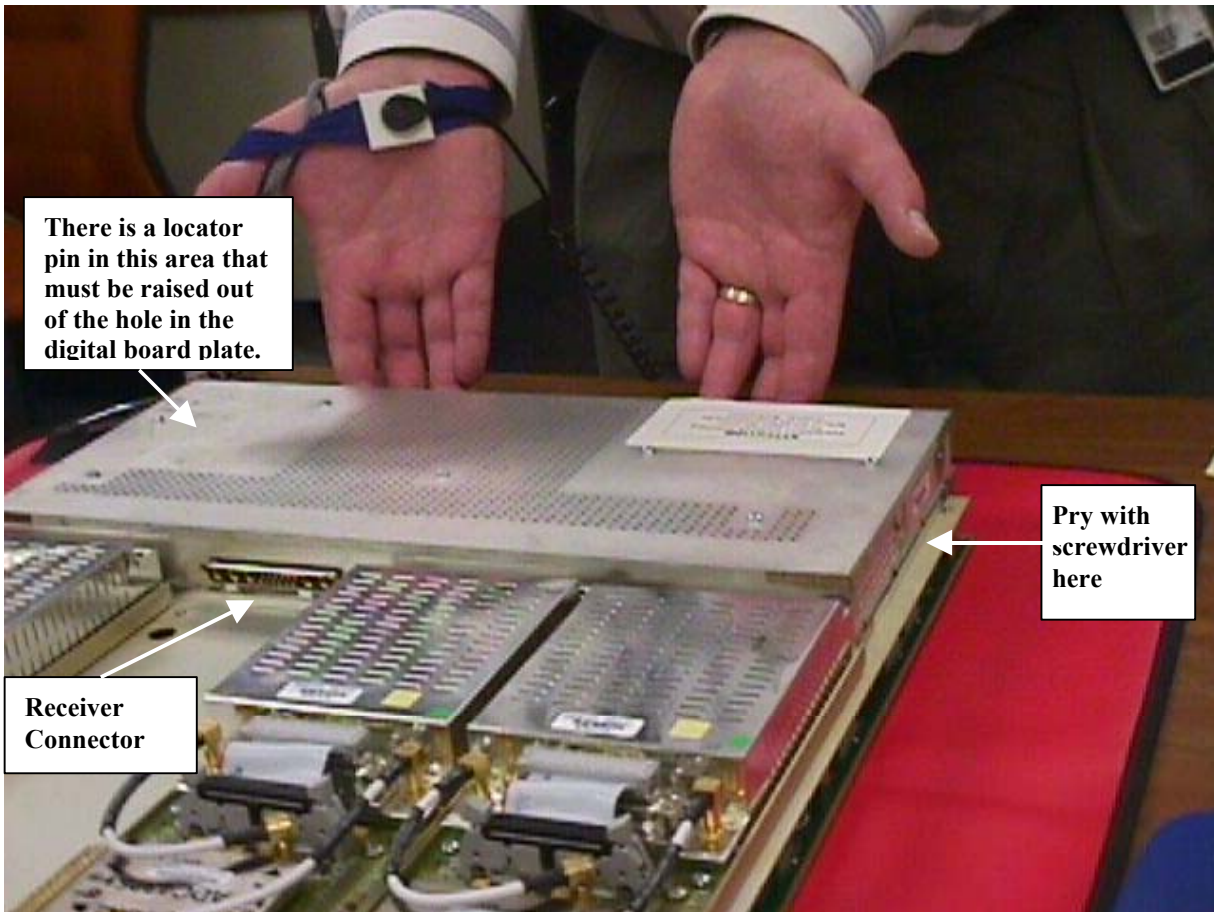


REMOVAL OF 2 RECEIVER SCREWS
ILLUSTRATION 1-4

2. The removal of the Exciter module requires removal of the UCERD front face plate. See Section 1-2 for removal of front face plate.
3. Use a screwdriver to gently pry the Exciter from the Digital Board. Gently lift up on the front of the Exciter. Take care not raise the Exciter excessively, or damage to the Receiver connectors may occur. The Exciter should become free from the Digital Board and the Receivers. There is a single connector that joins the Exciter to the Digital Board. There is also an alignment pin that inserts into the Digital Board. See Illustration 1-5.

WARNING

USE CAUTION REMOVING THE EXCITER MODULE. LIFTING EXCESSIVELY CAN BEND THE PINS ON THE RECEIVER MODULES OR DIGITAL BOARD CONNECTOR.

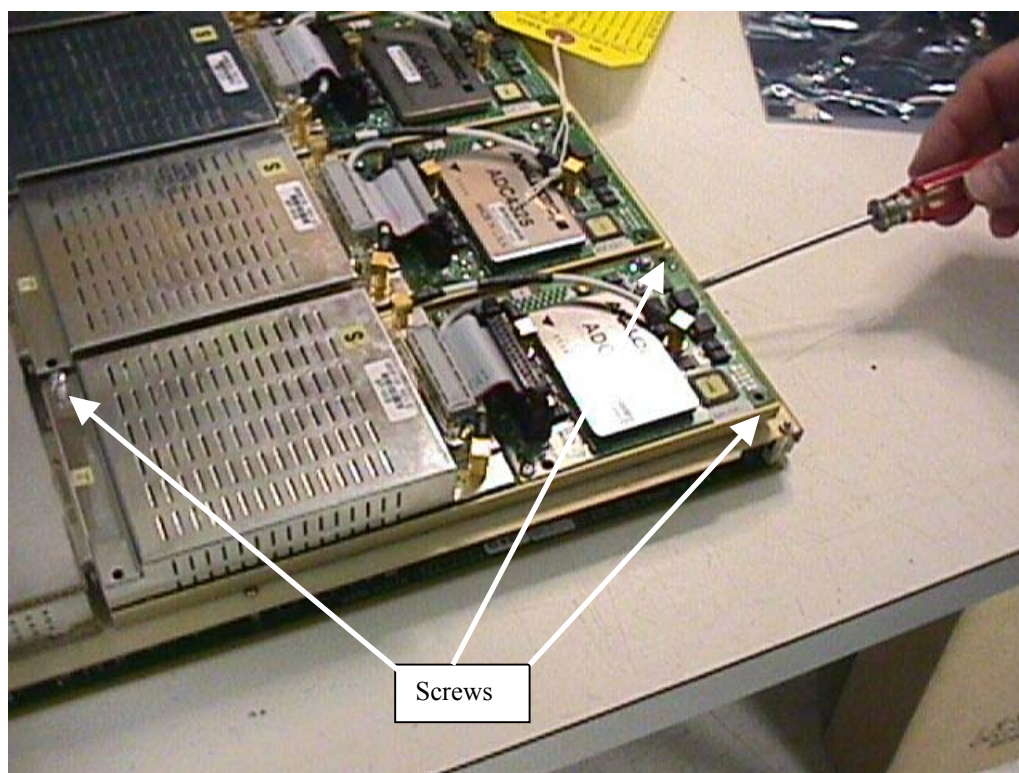


REMOVAL OF RF EXCITER FROM UCERD
ILLUSTRATION 1-5

4. To replace the Exciter, align the Receiver connections up and press firmly inward on the Exciter module until the connection with the Digital Board lines up. Once all the Receivers are full seated, press down on the Exciter to seat the connection to the Digital Board. See Section 1-2-2 to re-install the front face plate. Make sure EMC fingers are outside green protector cover (Illustration 1-3).

1-4 Replacement of Receiver Module(s)

1. The Receiver Modules are held in place by 2 screws. Receivers 1 and 4 have an addition screw located near the Exciter connection. Remove the 2 screws (or 3 screws for Receiver 1 or 4) and pry up on the receiver module with a screwdriver to remove it from the Digital Board. See illustration 1-6.

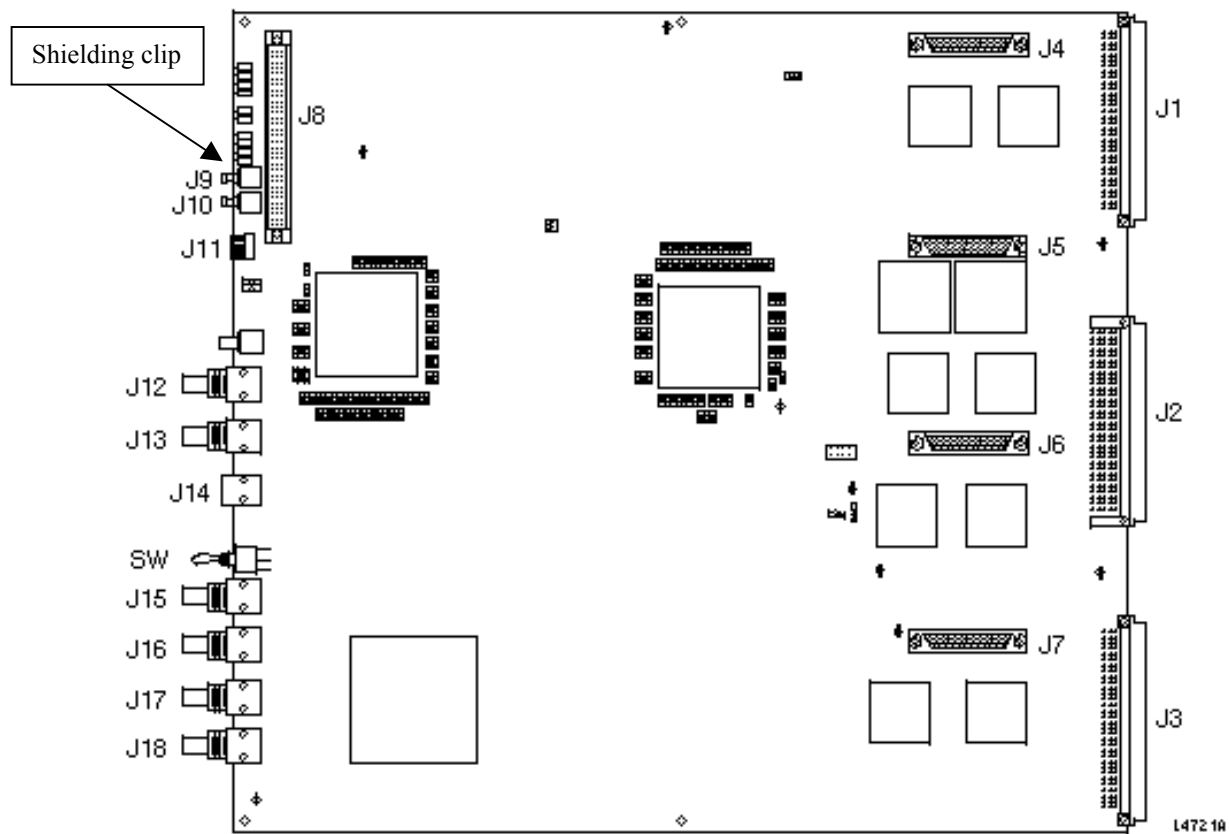


RECEIVER REMOVAL
ILLUSTRATION 1-6

2. To replace the Receiver, insert it into the Exciter module, then press down firmly on the Receiver module to seat the connection to the Digital board.
3. Install and tighten the 2 or 3 screws in Illustration 1-6.

1-5 Replacement of the Digital Board

1. The Digital Board provides support for the entire UCERD assembly. For removal, complete sections 1-1, 1-2, 1-3, and 1-4. Illustration 1-7 shows a Digital board without the protector covers.



DIGITAL BOARD ASSEMBLY
ILLUSTRATION 1-7

2. For installation, simply install Receivers, Exciter and face plate. Sections 1-4, 1-3, and 1-2, respectively.

Note

Make sure the shielding clip is attached to J9 and J10 on the front of the digital board. See illustration 1-8 for a picture of the clip.

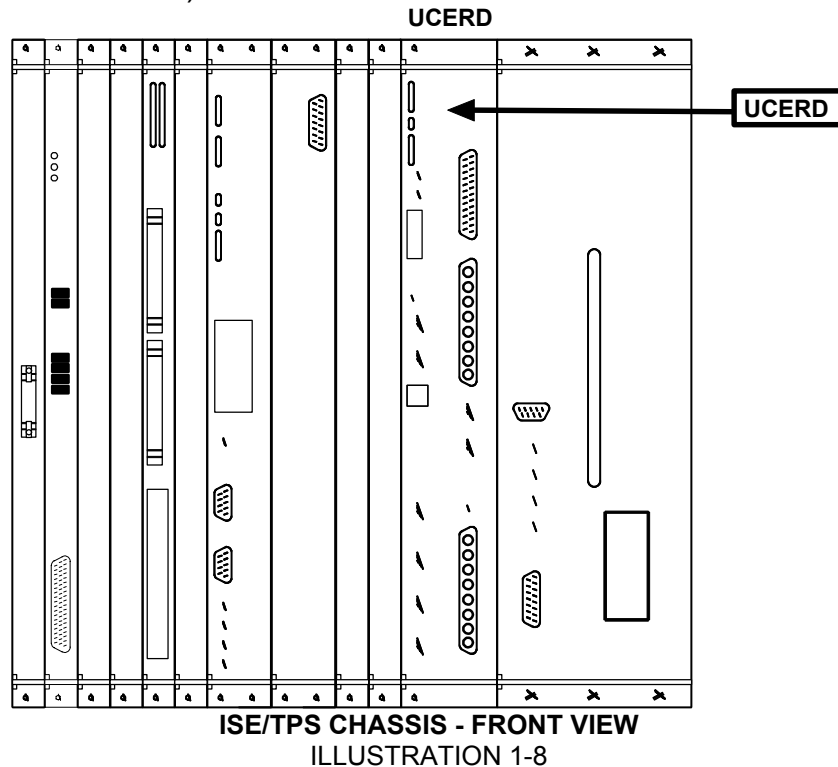


DIGITAL BOARD SHIELDING CLIP
ILLUSTRATION 1-8

3. The jumpers on the Digital board come preset. The jumpers are not to be moved in the field due to the fact that it requires removal of the cover plate, which is not to be done in the field.

1-6 Installation of UCERD Assembly Board

1. Install the appropriate module(s) into the UCERD Assembly (refer to sections 1-2, 1-3, 1-4)
2. Install the UCERD Assembly in the ISE/TPS Chassis and tighten the two front panel screws (see Illustration 1-8).



3. Turn on the main circuit breaker, CB1 on the System Cabinet.
4. Install the System Cabinet front cover.
5. If Signa software is running, **[Reset TPS]**.

1-7 Functional Checks Required

1. Allow thirty minutes for the RF Exciter and Receivers to warm up.
2. Using the Scope calibrator from the RF Power Measurement Kit, calibrate the scope for 8 divisions. (Refer to Quick Reference Card  from the RF Power Measurement Kit)
3. Disconnect J109 on the front panel of the UCERD. Connect cable 46-301549P6 (from TPS/RF cable kit) to the EXC RF OUT on UCERD.
4. Connect the BNC end of cable 46-301549P6 (from step 3) to the oscilloscope 50-ohm input. For high Z oscilloscope input, use the 50 ohm feed-through terminator supplied with the RF Power Measurement kit.

NOTE

If the oscilloscope has a 50 Ω selectable termination input, do not use the 50 Ω feed-through from the RF Power Measurement kit.

5. At the Operator Work Space, prepare the system for **APB Cal, series 1: Body** scan using the “Service Protocols” procedure located under the [Software Utilities] menu on the service methods CD-ROM, or for an alternate procedure, see the following.
 - a. Click on **[New Pt]**, and enter
 - Id: **geservice**
 - Name: **apb cal body**
 - Weight (Lb): **111**
 - Set Patient Protocols to **Service**.
 - b. In the protocol field, type **o.41.1** (o=Other, 41 =series) to load the APB CAL protocol.
6. Landmark on the table and **[Save Series]**.
7. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Setup Params]**. Set TG to **50**.
8. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Display CVs]**. Set value of CV **calmode** to **5** (Dual Logamp Waveform).
9. Set the value of **CV ia_rf1** to **32766** (sets 90° pulse full scale).
10. Set value of **CV ia_rf2** to **0** (turns off 180° pulse).
11. Select **[Research Operations]** , **[Download]**.
12. **[Manual Prescan]**. Set TG = **200**. Measure the Exciter output to make sure its peak RF power is at least 7.13 divisions on a calibrated oscilloscope with input termination of 50 ohms. (Refer to Table 1-2).

TABLE 1-2
UCERD SPECIFICATIONS

	Specification
Measurement Point	EXC RF OUT > 3dBm
EXC RF OUT	Greater than 7.13 Divisions (3dBm)

Note

The specifications are given in scope divisions, not volts. The scope needs to be calibrated with the scope calibrator in the RF Power Measurement kit when using an oscilloscope less than 150 MHz. If using a 300 MHz or greater oscilloscope, just measure the voltage. The minimum voltage for 3 dBm would be 893 mV_{pp}

13. Disconnect cable 46-301549p6 from the front of the Exciter.
14. Reconnect J109 at the front of the Exciter.
15. Perform procedure for EPI Bandpass Asymmetry Correction Characterization.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Oct. 14, 1999	R. Kaufman	Initial Version
1	Nov. 10, 1999	R. Kaufman	Procedural changes provided from Eng and Manufacturing.
2	June 27,2003	Hawthorne	Added step in 1-7 functional checks to do a BACC after new receiver installation.